THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 1

Chapter 5: **Circles** | Class 9 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** 48	Chord $= 2\sqrt{r^2 - d^2}$.
2	**(d)** 15	$d = \sqrt{r^2 - (\text{half-chord})^2}$.
3	**(a)** 37°	Half of the central angle.
4	**(c)** 147°	Angles on the two arcs are supplementary: $180^\circ - t$.
5	**(b)** 55°	The exterior angle equals the interior opposite angle.
6	**(c)** 50°	$OA = OB$ (radii) so the base angles are equal.
7	**(a)** equidistant	Recall the theorem.
8	**(b)** 14	Chord $= 2\sqrt{r^2 - d^2}$.
9	**(d)** 20	$d = \sqrt{r^2 - (\text{half-chord})^2}$.
10	**(a)** 44°	Half of the central angle.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Half-chord $= \sqrt{26^2 - 10^2} = 24$ cm (1); Chord $= 48$ cm (1).	1+1
12	Angle on the major arc $= 30^\circ$ (1); Angle on the minor arc $= 180^\circ - 30^\circ = 150^\circ$ (1).	1+1
13	Exterior angle = interior opposite angle (1); $\angle DCE = 62^\circ$ (1).	1+1
14	Half-chord $= \sqrt{26^2 - 24^2} = 10$ cm (1); Chord $= 20$ cm (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	$OA = OB$ so $\angle OAB = \angle OBA$ (1); $\angle OAB = \frac{180^\circ - 70^\circ}{2} = 55^\circ$ (1); Angle on the major arc $= \frac{70^\circ}{2} = 35^\circ$ (1).	1+1+1
16	$\angle A + \angle C = 180^\circ$; $5x + 5 = 180$ (1); $x = 35$ (1); $\angle A = 80^\circ$ (1).	1+1+1
17	$r = \sqrt{15^2 + 8^2} = 17$ cm; Right triangle legs 15 cm and 8 cm (1); Area $= \frac{1}{2} \times 15 \times 8$ (1); $= 60$ cm^2 (1).	1+1+1
	OR: $OA = OB$ so $\angle OAB = \angle OBA$ (1); $\angle OAB = \frac{180^\circ - 90^\circ}{2} = 45^\circ$ (1); Angle on the major arc $= \frac{90^\circ}{2} = 45^\circ$ (1).	1+1+1
18	$\angle A + \angle C = 180^\circ$; $5x + 5 = 180$ (1); $x = 35$ (1); $\angle A = 85^\circ$ (1).	1+1+1
	OR: $r = \sqrt{8^2 + 15^2} = 17$ cm; Right triangle legs 8 cm and 15 cm (1); Area $= \frac{1}{2} \times 8 \times 15$ (1); $= 60$ cm^2 (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Distance of first chord from centre $= \sqrt{13^2 - 5^2} = 12$ cm (1); Distance of second chord $= \sqrt{13^2 - 12^2} = 5$ cm (1); Same side: subtract the distances (1); Distance between chords $= \lvert 12 - 5 \rvert = 7$ cm (2).	1+1+1+2
	OR: $\angle C = 180^\circ - 75^\circ = 105^\circ$ (1); $\angle D = 180^\circ - 88^\circ = 92^\circ$ (1); $\angle DCE = \angle A = 75^\circ$ (1); $\angle DAF = 180^\circ - 75^\circ = 105^\circ$ (1); Reasons: opposite angles supplementary; exterior angle property (1).	1+1+1+1+1
20	Distance of first chord from centre $= \sqrt{17^2 - 8^2} = 15$ cm (1); Distance of second chord $= \sqrt{17^2 - 15^2} = 8$ cm (1); Same side: subtract the distances (1); Distance between chords $= \lvert 15 - 8 \rvert = 7$ cm (2).	1+1+1+2
	OR: $\angle C = 180^\circ - 66^\circ = 114^\circ$ (1); $\angle D = 180^\circ - 94^\circ = 86^\circ$ (1); $\angle DCE = \angle A = 66^\circ$ (1); $\angle DAF = 180^\circ - 66^\circ = 114^\circ$ (1); Reasons: opposite angles supplementary; exterior angle property (1).	1+1+1+1+1

Mistakes to avoid:

- Using the full chord instead of the half-chord in $r^2 = d^2 + (\text{half-chord})^2$.
- Forgetting that the perpendicular distance, not the slant distance, is used when finding the distance of a chord from the centre.
- Applying the angle theorems to an arc without checking the angle is at the *centre* (double) or on the *remaining* circle (half).
- For two parallel chords, adding the distances when they are on the same side or subtracting them when they are on opposite sides (the reverse is correct).
- Assuming that opposite angles of a cyclic quadrilateral are equal; they are supplementary (sum 180°).
- Using the angle in a semicircle theorem when the chord is not a diameter.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 2

Chapter 5: **Circles** | Class 9 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** 131°	Angles on the two arcs are supplementary: $180^\circ - t$.
2	**(d)** 68°	The exterior angle equals the interior opposite angle.
3	**(a)** 40°	$OA = OB$ (radii) so the base angles are equal.
4	**(d)** perpendicular to the chord	Recall the theorem.
5	**(c)** 40	Chord $= 2\sqrt{r^2 - d^2}$.
6	**(c)** 10	$d = \sqrt{r^2 - (\text{half-chord})^2}$.
7	**(b)** 56°	Half of the central angle.
8	**(a)** 122°	Angles on the two arcs are supplementary: $180^\circ - t$.
9	**(d)** 84°	The exterior angle equals the interior opposite angle.
10	**(c)** 30°	$OA = OB$ (radii) so the base angles are equal.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Angle on the major arc $= 45^\circ$ (1); Angle on the minor arc $= 180^\circ - 45^\circ = 135^\circ$ (1).	1+1
12	Exterior angle = interior opposite angle (1); $\angle DCE = 74^\circ$ (1).	1+1
13	Half-chord $= \sqrt{29^2 - 20^2} = 21$ cm (1); Chord $= 42$ cm (1).	1+1
14	Angle on the major arc $= 50^\circ$ (1); Angle on the minor arc $= 180^\circ - 50^\circ = 130^\circ$ (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	$OA = OB$ so $\angle OAB = \angle OBA$ (1); $\angle OAB = \frac{180^\circ - 110^\circ}{2} = 35^\circ$ (1); Angle on the major arc $= \frac{110^\circ}{2} = 55^\circ$ (1).	1+1+1
16	$\angle A + \angle C = 180^\circ$; $5x + 10 = 180$ (1); $x = 34$ (1); $\angle A = 88^\circ$ (1).	1+1+1
17	$r = \sqrt{16^2 + 12^2} = 20$ cm; Right triangle legs 16 cm and 12 cm (1); Area $= \frac{1}{2} \times 16 \times 12$ (1); $= 96$ cm^2 (1).	1+1+1
	OR: $OA = OB$ so $\angle OAB = \angle OBA$ (1); $\angle OAB = \frac{180^\circ - 130^\circ}{2} = 25^\circ$ (1); Angle on the major arc $= \frac{130^\circ}{2} = 65^\circ$ (1).	1+1+1
18	$\angle A + \angle C = 180^\circ$; $5x + 0 = 180$ (1); $x = 36$ (1); $\angle A = 77^\circ$ (1).	1+1+1
	OR: $r = \sqrt{12^2 + 16^2} = 20$ cm; Right triangle legs 12 cm and 16 cm (1); Area $= \frac{1}{2} \times 12 \times 16$ (1); $= 96$ cm^2 (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Distance of first chord from centre $= \sqrt{25^2 - 7^2} = 24$ cm (1); Distance of second chord $= \sqrt{25^2 - 24^2} = 7$ cm (1); Same side: subtract the distances (1); Distance between chords $= \lvert 24 - 7 \rvert = 17$ cm (2).	1+1+1+2
	OR: $\angle C = 180^\circ - 82^\circ = 98^\circ$ (1); $\angle D = 180^\circ - 71^\circ = 109^\circ$ (1); $\angle DCE = \angle A = 82^\circ$ (1); $\angle DAF = 180^\circ - 82^\circ = 98^\circ$ (1); Reasons: opposite angles supplementary; exterior angle property (1).	1+1+1+1+1
20	Distance of first chord from centre $= \sqrt{25^2 - 15^2} = 20$ cm (1); Distance of second chord $= \sqrt{25^2 - 20^2} = 15$ cm (1); Same side: subtract the distances (1); Distance between chords $= \lvert 20 - 15 \rvert = 5$ cm (2).	1+1+1+2
	OR: $\angle C = 180^\circ - 58^\circ = 122^\circ$ (1); $\angle D = 180^\circ - 103^\circ = 77^\circ$ (1); $\angle DCE = \angle A = 58^\circ$ (1); $\angle DAF = 180^\circ - 58^\circ = 122^\circ$ (1); Reasons: opposite angles supplementary; exterior angle property (1).	1+1+1+1+1

Mistakes to avoid:

- Using the full chord instead of the half-chord in $r^2 = d^2 + (\text{half-chord})^2$.
- Forgetting that the perpendicular distance, not the slant distance, is used when finding the distance of a chord from the centre.
- Applying the angle theorems to an arc without checking the angle is at the *centre* (double) or on the *remaining* circle (half).
- For two parallel chords, adding the distances when they are on the same side or subtracting them when they are on opposite sides (the reverse is correct).
- Assuming that opposite angles of a cyclic quadrilateral are equal; they are supplementary (sum 180°).
- Using the angle in a semicircle theorem when the chord is not a diameter.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 3

Chapter 5: **Circles** | Class 9 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(b)** supplementary	Recall the theorem.
2	**(a)** 30	Chord $= 2\sqrt{r^2 - d^2}$.
3	**(a)** 24	$d = \sqrt{r^2 - (\text{half-chord})^2}$.
4	**(d)** 68°	Half of the central angle.
5	**(b)** 108°	Angles on the two arcs are supplementary: $180^\circ - t$.
6	**(c)** 97°	The exterior angle equals the interior opposite angle.
7	**(a)** 20°	$OA = OB$ (radii) so the base angles are equal.
8	**(c)** equal	Recall the theorem.
9	**(c)** 48	Chord $= 2\sqrt{r^2 - d^2}$.
10	**(a)** 20	$d = \sqrt{r^2 - (\text{half-chord})^2}$.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Exterior angle = interior opposite angle (1); $\angle DCE = 91^\circ$ (1).	1+1
12	Half-chord $= \sqrt{29^2 - 21^2} = 20$ cm (1); Chord $= 40$ cm (1).	1+1
13	Angle on the major arc $= 65^\circ$ (1); Angle on the minor arc $= 180^\circ - 65^\circ = 115^\circ$ (1).	1+1
14	Exterior angle = interior opposite angle (1); $\angle DCE = 109^\circ$ (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	$OA = OB$ so $\angle OAB = \angle OBA$ (1); $\angle OAB = \frac{180^\circ - 150^\circ}{2} = 15^\circ$ (1); Angle on the major arc $= \frac{150^\circ}{2} = 75^\circ$ (1).	1+1+1
16	$\angle A + \angle C = 180^\circ$; $5x + 5 = 180$ (1); $x = 35$ (1); $\angle A = 82^\circ$ (1).	1+1+1
17	$r = \sqrt{24^2 + 7^2} = 25$ cm; Right triangle legs 24 cm and 7 cm (1); Area $= \frac{1}{2} \times 24 \times 7$ (1); $= 84$ cm^2 (1).	1+1+1
	OR: $OA = OB$ so $\angle OAB = \angle OBA$ (1); $\angle OAB = \frac{180^\circ - 50^\circ}{2} = 65^\circ$ (1); Angle on the major arc $= \frac{50^\circ}{2} = 25^\circ$ (1).	1+1+1
18	$\angle A + \angle C = 180^\circ$; $5x + 5 = 180$ (1); $x = 35$ (1); $\angle A = 78^\circ$ (1).	1+1+1
	OR: $r = \sqrt{7^2 + 24^2} = 25$ cm; Right triangle legs 7 cm and 24 cm (1); Area $= \frac{1}{2} \times 7 \times 24$ (1); $= 84$ cm^2 (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Distance of first chord from centre $= \sqrt{10^2 - 6^2} = 8$ cm (1); Distance of second chord $= \sqrt{10^2 - 8^2} = 6$ cm (1); Same side: subtract the distances (1); Distance between chords $= \lvert 8 - 6 \rvert = 2$ cm (2).	1+1+1+2
	OR: $\angle C = 180^\circ - 97^\circ = 83^\circ$ (1); $\angle D = 180^\circ - 64^\circ = 116^\circ$ (1); $\angle DCE = \angle A = 97^\circ$ (1); $\angle DAF = 180^\circ - 97^\circ = 83^\circ$ (1); Reasons: opposite angles supplementary; exterior angle property (1).	1+1+1+1+1
20	Distance of first chord from centre $= \sqrt{5^2 - 3^2} = 4$ cm (1); Distance of second chord $= \sqrt{5^2 - 4^2} = 3$ cm (1); Same side: subtract the distances (1); Distance between chords $= \lvert 4 - 3 \rvert = 1$ cm (2).	1+1+1+2
	OR: $\angle C = 180^\circ - 112^\circ = 68^\circ$ (1); $\angle D = 180^\circ - 76^\circ = 104^\circ$ (1); $\angle DCE = \angle A = 112^\circ$ (1); $\angle DAF = 180^\circ - 112^\circ = 68^\circ$ (1); Reasons: opposite angles supplementary; exterior angle property (1).	1+1+1+1+1

Mistakes to avoid:

- Using the full chord instead of the half-chord in $r^2 = d^2 + (\text{half-chord})^2$.
- Forgetting that the perpendicular distance, not the slant distance, is used when finding the distance of a chord from the centre.
- Applying the angle theorems to an arc without checking the angle is at the *centre* (double) or on the *remaining* circle (half).
- For two parallel chords, adding the distances when they are on the same side or subtracting them when they are on opposite sides (the reverse is correct).
- Assuming that opposite angles of a cyclic quadrilateral are equal; they are supplementary (sum 180°).
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